Code No: 157EQ **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, January/February - 2023
FUNDAMENTALS OF BIOMEDICAL APPLICATIONS
(Common to CSE, IT)

Time: 3 Hours **Max.Marks:75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	Give the Nernst equation for electrode potential.	[2]
b)	Differentiate between action potential and resting potential.	[3]
c)	Name the parts of the heart conduction system.	[2]
d)	What are the requirements of physiological signal amplifier?	[3]
e)	What is korotkoft sound?	[2]
f)	Discuss about the origin of heart sounds.	[3]
g)	Differentiate between external pacemaker and implanted pacemaker.	[2]
h)	What are the advantages diathermies?	[3]
i)	Summarize latency in EMG.	[2]
j)	Give the frequency range & amplitude of EEG and EMG waves.	[3]

PART – B

(50 Marks)

2.a) Sketch the block diagram of biomedical instrumentation system and explain the function of each block.
b) Draw the structure of a living cell of a body and explain its constituents. [5+5]

OR

3.a) With help of neat diagram, explain the electrical activity associated with one contraction of muscles.
b) Name the different propagation methods of action potentials. [5+5]

4.a) Justify how bipolar leads can be used for measurement of ECG with help of neat diagrams
b) Enumerate and explain the different ECG waves and intervals. [5+5]

OR

5.a) Illustrate the working of the cardiac conduction path with neat diagram.
b) Explain the working principle of a ECG machine with a neat block diagram. [5+5]

6.a) Describe the working principle of ultrasonic blood pressure measurement.
b) Explain the measurement of heart sound with suitable diagram. [5+5]

OR

7.a) Justify and explain how electromagnetic principle can be used for measurement of blood flow with help of neat diagram**.**
b) Discuss the Oscillometric blood pressure1 measurement method. [5+5]

8.a) Explain the basic principle of operation of an ultrasonic diathermy unit.
b) With help of neat block diagram, explain the working of hemodialysis machine. [5+5]

OR

9.a) Explain the working principle of surgical diathermy unit with a neat block diagram.
b) What is a Synchronized defibrillator? Draw a block diagram of it and explain its working. [5+5]

10.a) Discuss in brief about any one type of pneumotachmeter with relevant diagram**.**
b) With a neat diagram, explain the working principle of EMG. [5+5]

OR

11.a) What is EMG nerve conduction velocity? Explain the process and purpose of EMG NCV test.
b) Explain the different modes of ventilator. [5+5]

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